Workflow Orchestration and Deployment

Thursday, January 29, 2026 5:57 PM

Recap: Experiment tracking and Model Registry

- Track all runs → HPT → For any algo The Influ on hyper param on perf of algo.
  - ML Flow
  - KNN, DT → add plot → parallel Coord plot
- Register Model → Save model → Versioning
- Select best perf model
  - Criterions
    1) Generalization Tr Tes < 5%
    2) Train
    3) Test
    4) Size model
  - Manually → ML flow, Log metric → Parallel Coord plot → Made decision → Suitable / Best model for Deployment

Agenda

10-15 ⇒ Workflow Orchestration – Prefect

Selected a best model → Iris data → 266 Run

Model → EC2 → Host Model on EC2 Inst → End point → user ask query.

Workflow Orchestration

What? → Manage different

Why

Prefect ← Early

Airflow

Kubeflow

ML flow

Manage different steps in a workflow

Observability. (Stages in a pipeline)

Automate & Schedule steps workflow

Kubeflow
MLflow

Manually perf Tedious / Mondany rep task

quite diff

We need some tools to automate rep jobs.

Internet / e Commerce
→ Amazon
→ a lot of customer
→ make / orders
→ Capture customer beh
→ [Recommend product]

Manually
→ a lot
→ man power

{Billions user} {every user}

Task: Based on obs customer beh
→ Rec product } Rep

→ Automate - task
→ Tool ⇒ prefect

Mont
→ Update prod / web.
→ odd times ⇒ During weekends

User Traffic less

{Updation happends automation}

Schedules
→ Make updated data available to server

{ → midnights
→ all team ⇒ Deploy / Update

⇒ prefect ⇒ Divide work flow
→ stages - Stage-1
Stage-2

Tasks ← Module ← Stage-n

Combine Task ⇒ Flow → Automate flow

Simple syntax

from dataset → pipeline → seq stages → Task ⇒ function(py)

Combine Tasks ⇒ flow → Automate flow